

Nitant Rai

ACADEMIC CV

nitant.raio7@gmail.com | github.com/nitantrai | linkedin.com/in/nitantrai

Google Scholar: https://scholar.google.com/citations?user=t3mQx_kAAAAJ

RESEARCH INTERESTS

Forest remote sensing, forest biometrics, UAV-based regeneration monitoring, LiDAR, computer vision, and reproducible geospatial analysis.

EDUCATION

Virginia Tech | 2023-present

Ph.D. Candidate in Forestry; M.Eng. in Computer Science, in progress. Dissertation: Remote Sensing and Machine Learning Applications for Forest Monitoring.

Mississippi State University | 2021-2022

M.S. in Forestry; minor in Geoscience. Thesis: Mapping Forest Structure in Mississippi Using Lidar Remote Sensing.

Tribhuvan University, Nepal | 2014-2018

B.S. in Forestry.

RESEARCH EXPERIENCE

Graduate Research Assistant | Virginia Tech | Jan 2023-present

Develop Python and R workflows integrating UAV imagery, LiDAR, and field inventory data for forest structure assessment and regeneration monitoring. Apply computer vision, statistical modeling, quality checks, and uncertainty analysis. Conduct UAV data collection and communicate findings through maps, reports, and presentations.

Graduate Research Assistant | Mississippi State University | 2021-2022

Processed airborne and spaceborne LiDAR, including ICESat-2, for canopy height and forest structure analysis. Evaluated uncertainty in remotely sensed measurements and delivered instruction on backpack LiDAR collection and processing.

TEACHING AND APPLIED EXPERIENCE

Teaching experience | Virginia Tech

Instruct students in ArcGIS Pro, map interpretation, forest inventory, vegetation assessment, and field measurement protocols. Support field-based learning in soil and water evaluation, silviculture, and forestry Best Management Practices.

Intern | Virginia Tech Drone Park | May-August 2025

Operated and maintained drones, developed flight plans, supported proficiency training and demonstrations, and explained research applications to students and visitors.

2026 | Journal article

S Rijal, S Sharma Banjade, N Rai, AK Mandal, R Parajuli. Improving Flood Detection in the Himalayas: Evaluating SAR-Optical Data Fusion with Machine Learning in a Mountainous River Basin of Nepal. *Journal of the Indian Society of Remote Sensing*, 1-14.

2025 | Journal article

J Oli, N Rai, R Sharma, N Baral. Forest fire trends and management practices in the hills of Nepal. *GIS Odyssey Journal* 5 (1), 189-202.

2025 | Journal article

N Baral, S Joshi, N Rai, AB Joshi, K Poudel, SK Yadav. Breeding habitat suitability modeling of white-rumped vulture (*Gyps bengalensis*, Gmelin JF, 1788) in proximity to vulture restaurants in central lowland of Nepal. *Frontiers in Conservation Science* 6, 1645493.

2025 | Preprint

S He, Z Osman, F Cladera, D Ong, N Rai, PC Green, V Kumar, P Chaudhari. Estimating the Diameter at Breast Height of Trees in a Forest from RGB. *arXiv:2505.03093* (2025; revised 2026).

2024 | Journal article

N Rai, Q Ma, KP Poudel, A Himes, Q Meng. Evaluating the uncertainties in forest canopy height measurements using ICESat-2 data. *Journal of Remote Sensing* 4, 0160.

2023 | Conference / proceedings

S Sharma Banjade, N Rai, B Subedi. Comparison of supervised classification algorithms using a hyperspectral image for land use/land cover classification. *Environmental Sciences Proceedings* 29 (1), 59.

2022 | Thesis

N Rai. Mapping forest structure in Mississippi using LiDAR remote sensing. Mississippi State University.

2021 | Journal article

S Joshi, N Rai, R Sharma, N Baral. Land Use/Land Cover (LULC) Change in Suburb of Central Himalayas: A Study from Chandragiri, Kathmandu. *Journal of Forest and Environmental Science* 37 (1), 44-51.

2020 | Conference / proceedings

N Rai, Q Ma, J Yang. Mapping Forest Height in Lower Mississippi Alluvial Valley. 41st Asian Conference on Remote Sensing (ACRS 2020).

SELECTED PRESENTATIONS

2026 | Advancing UAV-based Seedling Detection for Forest Regeneration Monitoring using CNN and Transformer Models. International Conference on Applications and Solutions in Digital Forestry, Purdue University.

2025 | Deep Learning Approaches for Detecting Tree Seedlings using UAV Imagery. SOMENS and NEMO Annual Meeting.

2025 | Advancing UAV Seedling Detection with Deep Learning. Forest Modeling Research Cooperative Annual Meeting.

UAV Seedling Detection and Monitoring Pipeline

Python and YOLO-based batch inference for seedling detection and counting, with CSV summaries and visual outputs. Repository: github.com/nitantrai/UAV_seedling_monitoring_pipeline

TECHNICAL SKILLS

Python, R, SQL, PyTorch, scikit-learn, NumPy, Pandas, GeoPandas, GDAL, rasterio, ArcGIS Pro, Google Earth Engine, Git/GitHub. Methods: LiDAR processing, UAV imagery, computer vision, statistical modeling, model validation, forest inventory.

SERVICE, CERTIFICATION, AND AWARDS

Peer review: Sustainable Geosciences: People, Planet, and Prosperity; Trees, Forests and People.

FAA Part 107 UAS Remote Pilot certification.

2021 | Second place, 67th Southern Hardwood Forest Research Group poster competition.

2021 | Second place, Graduate Student Research Symposium, Mississippi State University.

2019 | Govinda Narsing Rayamajhi Memorial Award, Nepal.

LANGUAGES

English, Nepali, Hindi.